

Project Title: Visualizing U.S. Natural Disaster
Declarations

Name: Supreetha J U

College: Sona College of Technology, Salem

Domain: Data Visualization

Milestone: 03 – Geographic Analysis

1. Objective

- The objective of this milestone is to analyze the geographical distribution of disaster declarations across different states in the United States.
- This analysis focuses on identifying disaster-prone regions, understanding regional variations, and examining how different types of disasters occur across locations. By using geographic visualizations, we aim to uncover patterns that are not easily visible in raw data.

2. Problem Statement

This milestone addresses the following key questions:

- Which states experience the highest number of disaster declarations?
- Are disasters concentrated in specific regions?
- Do certain disaster types occur more frequently in particular states?
- Can we identify geographic hotspots using visualization techniques?

Answering these questions helps in understanding the spatial behavior of disasters.

3. Dataset Used

The dataset used for this analysis is **usnd_cleaned.csv**, which was prepared during Milestone 01.

Columns Used

- state – Represents disaster location
- incidentType – Type of disaster
- year, month – Time-related features

Before performing analysis, the dataset was checked for missing values, consistency, and correctness to ensure reliable results.

4. Geographic Aggregation

4.1 State-Level Aggregation

The dataset was grouped by state to calculate the total number of disaster declarations for each state. This step helps in identifying which states are more frequently

affected by disasters and forms the foundation for further visualization.

```
#State-wise Aggregation
state_counts = df['state'].value_counts().rename_axis('state').reset_index(name='disaster_count')
```

✓ 0.2s

Generate Code Markdown

```
state_counts.head()
```

✓ 0.5s

	state	disaster_count
0	TX	3842
1	MO	2263
2	KY	2026
3	VA	1982
4	OK	1882

The above output shows the total number of disasters recorded for each state. It is observed that certain states have significantly higher counts compared to others, indicating uneven distribution of disasters.

4.2 State + Incident Type Aggregation

The dataset was further grouped by both state and incident type. This allows us to analyze how different types of disasters are distributed within each state. The below output highlights which disaster types are dominant in each state. It helps in understanding the relationship between geography and disaster type.

```
grouped_data = df.groupby(['state', 'incidentType']).size().reset_index(name='count')
grouped_data.head()
```

✓ 0.3s

	state	incidentType	count
0	AK	Earthquake	7
1	AK	Fire	16
2	AK	Flood	46
3	AK	Other	4
4	AK	Snow	4

4.3 Regional Aggregation

States were grouped into four major regions:

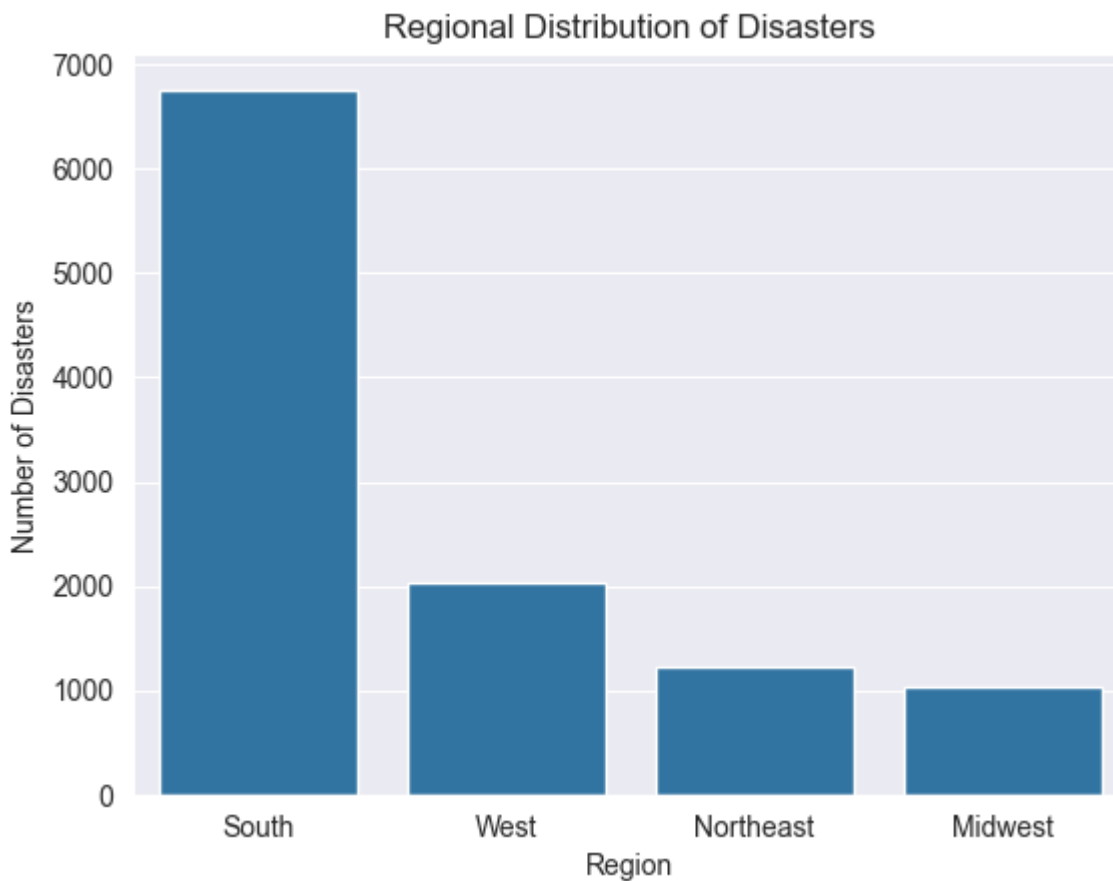
- West
- Midwest
- South
- Northeast

This grouping helps in comparing disaster occurrences at a broader level.

5. Visualizations

5.1 Regional Comparison Chart

A bar chart was created to compare disaster counts across different regions.



Interpretation

The South region shows the highest number of disaster declarations, mainly due to hurricanes and floods. The West also shows high values due to wildfires. The Midwest and Northeast have comparatively lower disaster counts.

5.2 Choropleth Map – Disaster Distribution

A choropleth map was used to visualize disaster distribution across states.

Geographical Spread of Disaster Declarations



Interpretation

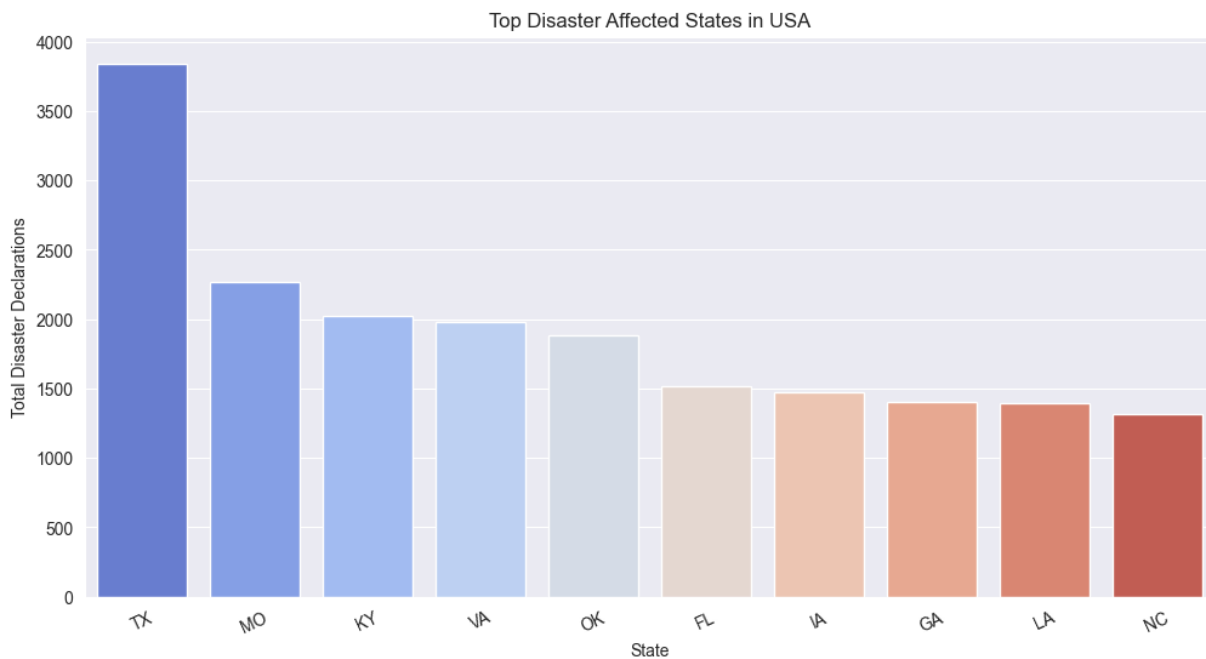
States with darker shades represent higher disaster frequencies. Coastal and southern states such as Texas and Florida show higher disaster counts, indicating higher vulnerability.

5.3 Top 10 States Comparison

A bar chart was created to display the top 10 states with the highest disaster counts.

Interpretation

The chart shows that a small number of states contribute to a large portion of total disaster declarations. This highlights geographic imbalance.

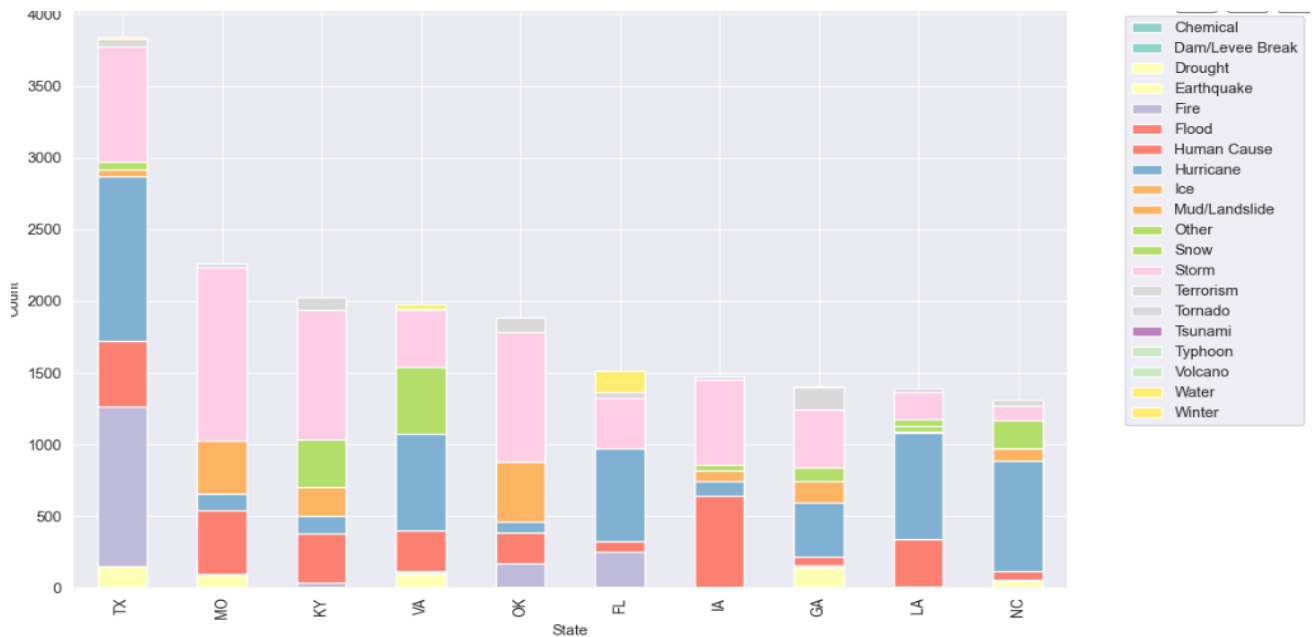


5.4 Incident Type Distribution by State

A stacked bar chart was created to show how different disaster types contribute within each state.

Interpretation

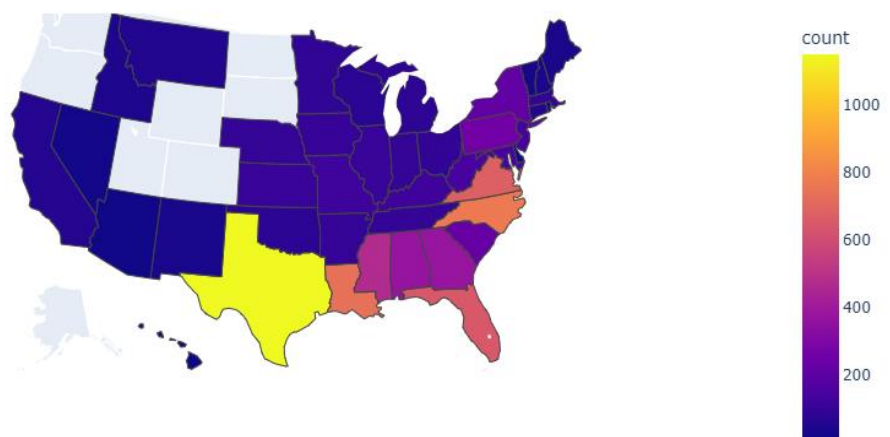
Different states show different dominant disaster types. For example, wildfires are more common in western states, while hurricanes are more frequent in coastal areas.



5.5 Hurricane Hotspot Map

A map was created to highlight hurricane-prone regions.

Hurricane Disaster Hotspots



Interpretation

The map clearly shows that coastal regions, especially near the Gulf of Mexico and Atlantic Ocean, experience a higher number of hurricane-related disasters.

6. Tools and Libraries Used

- **Pandas** – Data cleaning and aggregation
- **NumPy** – Numerical computations
- **Matplotlib** – Basic plotting
- **Seaborn** – Advanced visualizations
- **Plotly** – Interactive maps

7. Key Insights

- Disaster occurrences are unevenly distributed across states
- Southern and coastal regions are more disaster-prone
- Disaster types vary depending on geographic conditions
- Visualization helps in identifying patterns and hotspots effectively

8. Conclusion

This milestone successfully analyzed the geographic distribution of disaster declarations. By using aggregation and visualization techniques, important patterns such as high-risk states and regional trends were identified.

The analysis highlights how geography and climate influence disaster occurrences, making geographic visualization an essential tool in disaster analysis.